HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026 · PHASE 1

GlycoTwin

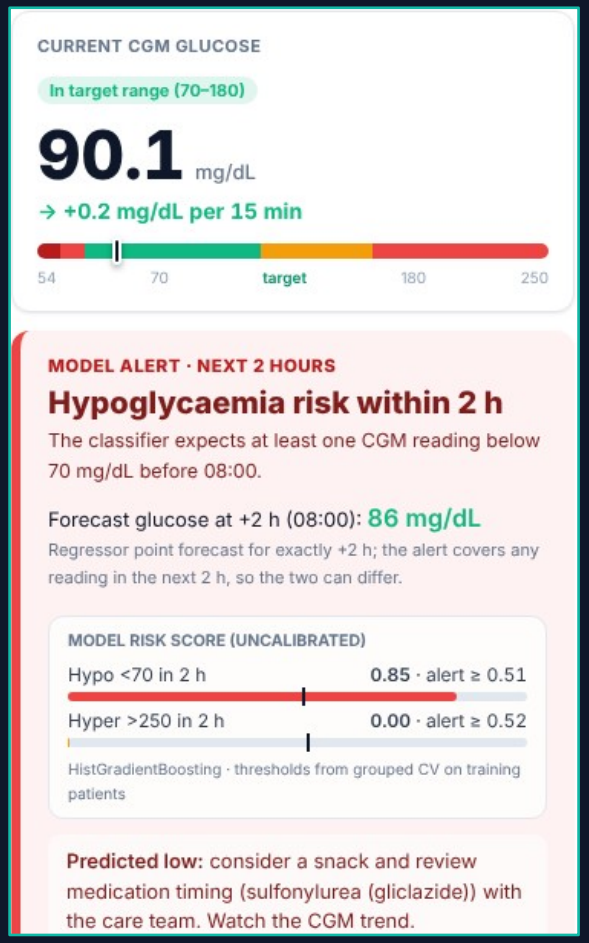
A multimodal digital twin for Type 2 diabetes that forecasts hypoglycaemia (<70 mg/dL) and marked hyperglycaemia (>250 mg/dL) 2 hours ahead, by fusing a synthetic EHR with synthetic CGM and wearable data

Team Classy Leopards

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Held-out patient: CGM in range now, hypoglycaemia alert for the next 2 h

Synthetic data only · research prototype, not a medical device · not for dosing decisions

THE PROBLEM IN INDIA

Diabetes is common in India, and dangerous glucose swings happen between clinic visits

101 million

people in India had diabetes in 2021 (11.4% of adults), counted by testing, including undiagnosed cases [1]

136 million

had prediabetes (15.3%) [1]

BMI 23.9 ≈ 30

South Asian adults reach the diabetes incidence of white adults with BMI 30 at a BMI of about 23.9 [2]

What goes wrong between visits

- **Hypoglycaemia:** Level 1 is <70 mg/dL, Level 2 is <54 mg/dL [3]. In Type 2 diabetes the main drug causes are sulfonylureas and insulin; risk rises with age and lower kidney function.
- **Marked hyperglycaemia:** >250 mg/dL. The time-in-range consensus asks for more than 70% of readings in 70–180 mg/dL and fewer than 4% below 70 [4].
- **Care is episodic:** clinic reviews happen every few months. A CGM shows the trend, but someone still has to notice a trend heading out of range and act before it gets there.

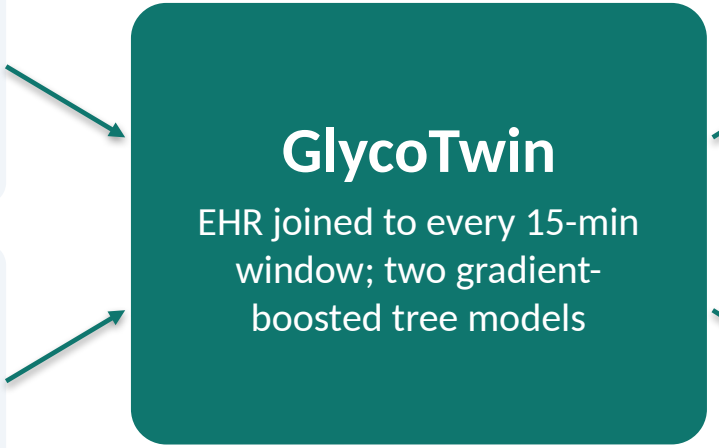
[1] Anjana RM et al., ICMR-INDIAB-17, Lancet Diabetes Endocrinol 2023 · [2] Caleyachetty R et al., Lancet Diabetes Endocrinol 2021
[3] ADA Standards of Care in Diabetes 2025, section 6 · [4] Battelino T et al., Diabetes Care 2019

OUR SOLUTION

Every 15 minutes, one fused feature set and two models answer two questions per patient

Static EHR
Demographics, labs, medication, history, TCF7L2 genotype

Live wearable stream
CGM, heart rate, HRV, steps, sleep stage, carbs, insulin



1. Event alert
Will any reading in the next 2 h be <70 or >250 mg/dL? Risk score vs a set threshold

2. Glucose forecast
What will the CGM read exactly 2 h from now (mg/dL)?

How a doctor uses the twin

Cohort risk board
patients ranked by current 2-h risk score

Live monitor
EHR card, last 8 h of data, forecast and alert

What-if sandbox
rule-based counselling tool, separate from the model

Model card
every number read from models/metrics.json

DATA: TWO SYNTHETIC STREAMS

Two synthetic streams for the same 200 people, aligned every 15 minutes

Static EHR data/synthetic_ehr.csv

200 personas

- Age, sex, diet pattern, BMI; HbA1c, fasting glucose, BP, eGFR, LDL, triglycerides
- Medication, insulin regimen, risk tier, past diagnoses
- Prior hypos (12 m), family history, TCF7L2 rs7903146 risk alleles

HbA1c 5.7–9.7% (mean 6.82%); 66 on a sulfonylurea, 3 on insulin

Live stream data/synthetic_wearables.csv

132,800 windows 200 × 664 labelled windows (7 days of 15 min data)

- CGM glucose, heart rate, HRV, steps, sleep stage (Awake/Light/Deep/REM)
- Logged meal carbohydrates and insulin units
- Labels: any reading <70 or >250 in the next 2 h; glucose exactly 2 h ahead

Windows: 128,086 no event · 1,012 hypo · 3,702 hyper. 409 readings <70 (35 <54) in 37 patients

The simulator couples the streams on purpose

- Sulfonylurea taken on a skipped breakfast causes lows; missed or extra insulin boluses
- Low glucose raises heart rate and lowers HRV; short sleep raises next-day insulin resistance
- TCF7L2 risk alleles raise post-meal peaks; eGFR <60 slows insulin clearance (no persona is below 60, so not exercised)

Caveat

Parameters are illustrative and not calibrated to any dataset. Population ranges are loosely informed by published Indian literature. Meal, walk and sleep times vary day to day.

Source: data/synthetic_*.csv, models/metrics.json · Synthetic data only · research prototype, not a medical device · not for dosing decisions

MODEL

Two gradient-boosted tree models on 34 fused features

34 features = 18 wearable + 16 EHR

All computed from the current and past windows only

- **5** **CGM** current glucose, change over 15 and 30 min, 1 h mean and SD
- **4** **Heart** heart rate and HRV, now and 1 h means
- **3** **Activity** steps in the last 15 min, 1 h and 2 h
- **4** **Meals, insulin** carbohydrates and insulin units, now and 2 h sums
- **2** **Context** hour of day, sleep stage
- **16** **EHR** age, sex, BMI, diabetes duration, HbA1c, fasting glucose, BP, eGFR, diet, medication, insulin regimen, risk tier, prior hypos, family history, TCF7L2

Two scikit-learn HistGradientBoosting models

- Classifier: 3 classes (no event / <70 / >250 in the next 2 h), class_weight="balanced" because events are rare
- Regressor: CGM glucose exactly 2 h ahead
- Max depth 6, learning rate 0.08, up to 150 iterations, native categorical splits

Split by patient: tested on unseen people

- 150 training / 50 held-out patients, none in both; held-out = 33,200 windows never used for training or thresholds

Alert rule

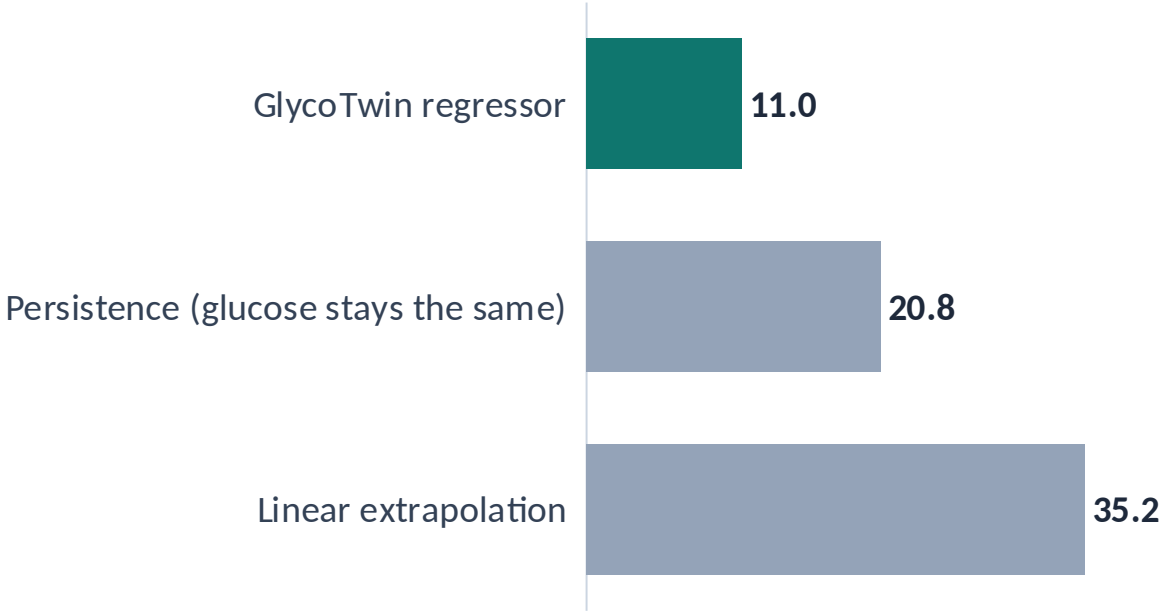
- Hypo alert if score ≥ 0.51, else hyper alert if score ≥ 0.52; thresholds maximise F2 on 5-fold patient-grouped CV of the training patients
- Scores are uncalibrated risk scores, not probabilities. About 15 ms per window on CPU

Source: models/train_fusion_model.py, models/metrics.json · Synthetic data only · research prototype, not a medical device · not for dosing decisions

RESULTS · 2-HOUR GLUCOSE FORECAST

On 50 unseen patients the 2-hour forecast error is 11.0 mg/dL, 1.9× lower than assuming glucose stays the same

Mean absolute error at +2 h (mg/dL, lower is better)



R^2 0.803 vs 0.428 for persistence. 5-fold patient-grouped CV over all 200 patients: MAE 11.08 mg/dL, R^2 0.806.

Clarke error grid zones (% of forecasts)

Model	A	B	C	D	E
GlycoTwin	**91.8%**	**7.6%**	**0.0%**	**0.6%**	**0.0%**
Persistence	79.1%	19.7%	0.0%	1.2%	0.0%
Linear extrap.	59.9%	34.4%	4.1%	1.4%	0.2%

99.4% in zones A+B

91.81% A + 7.58% B. Zone D 0.61%; none in C or E

Each forecast is compared with the CGM reading exactly 2 h later. Zones per Clarke et al., Diabetes Care 1987.

RESULTS · 2-HOUR EVENT ALERTS

Alerts warned of 34 of 39 lows and 42 of 43 highs in advance, but most alarms are false

Hypoglycaemia (<70 mg/dL within 2 h)

34/39	73.4%	52.3%
episodes warned 15–120 min ahead	window sensitivity	window precision

- PR-AUC 0.621 · ROC-AUC 0.990 · median lead time 120 min
- If CGM is 70–250 now: sensitivity 63.2%, precision 39.3%
- Simple rule (CGM <90 and falling): sensitivity 42.8%, precision 33.9%
- **0.36 alarms per patient-day; 69% not followed by the event within 2 h**

Marked hyperglycaemia (>250 mg/dL within 2 h)

42/43	86.2%	36.4%
episodes warned 15–120 min ahead	window sensitivity	window precision

- PR-AUC 0.774 · ROC-AUC 0.988 · median lead time 120 min
- If CGM is 70–250 now: sensitivity 77.9%, precision 24.0%
- Simple rule (CGM >200 and rising): sensitivity 40.8%, precision 15.6%
- **0.58 alarms per patient-day; 88% not followed by the event within 2 h**

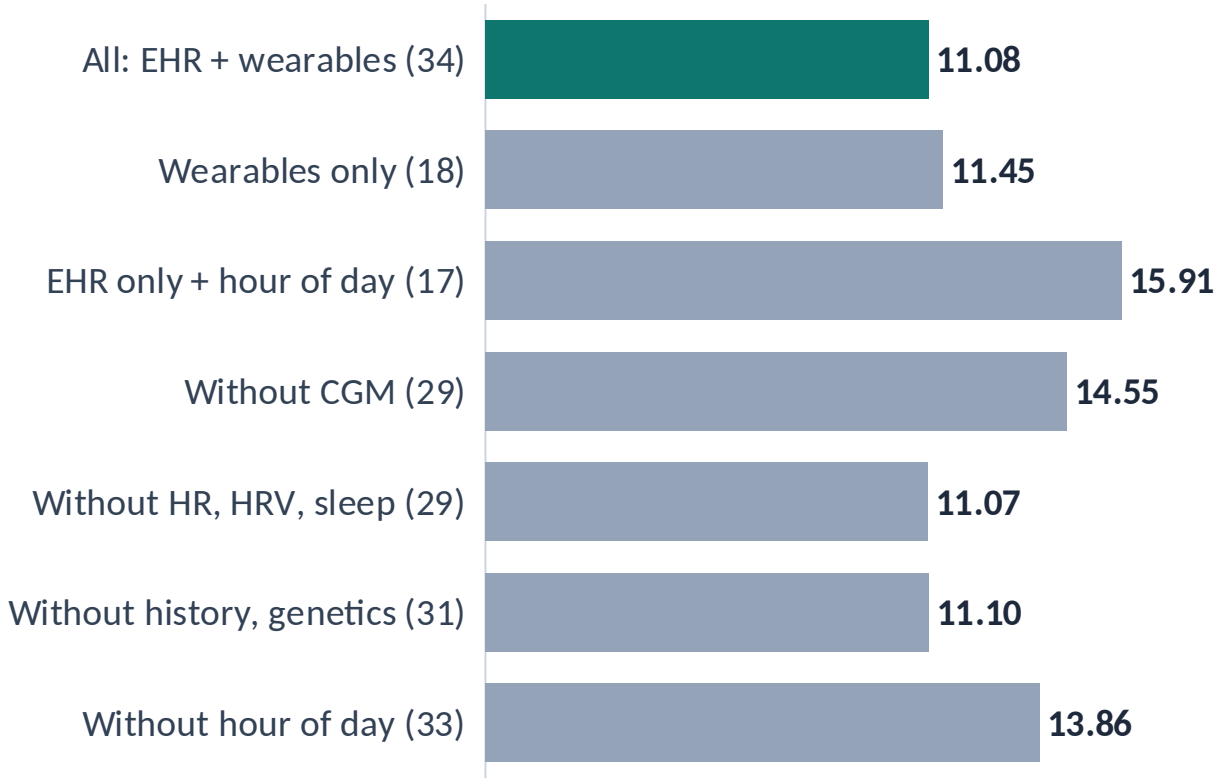
Honest reading: thresholds favour sensitivity (F2), so alarm fatigue is the main weakness to fix with clinicians. 50 held-out patients, 458 hypo and 557 hyper windows; scores are uncalibrated.

Source: models/metrics.json · Synthetic data only · research prototype, not a medical device · not for dosing decisions

FUSION EVIDENCE

Fusing the EHR helps modestly for the forecast and lows; highs are driven by the wearables

Forecast MAE by feature set (5-fold patient-grouped CV, mg/dL)



Lower is better; the shipped model uses all features.

EHR-swap test

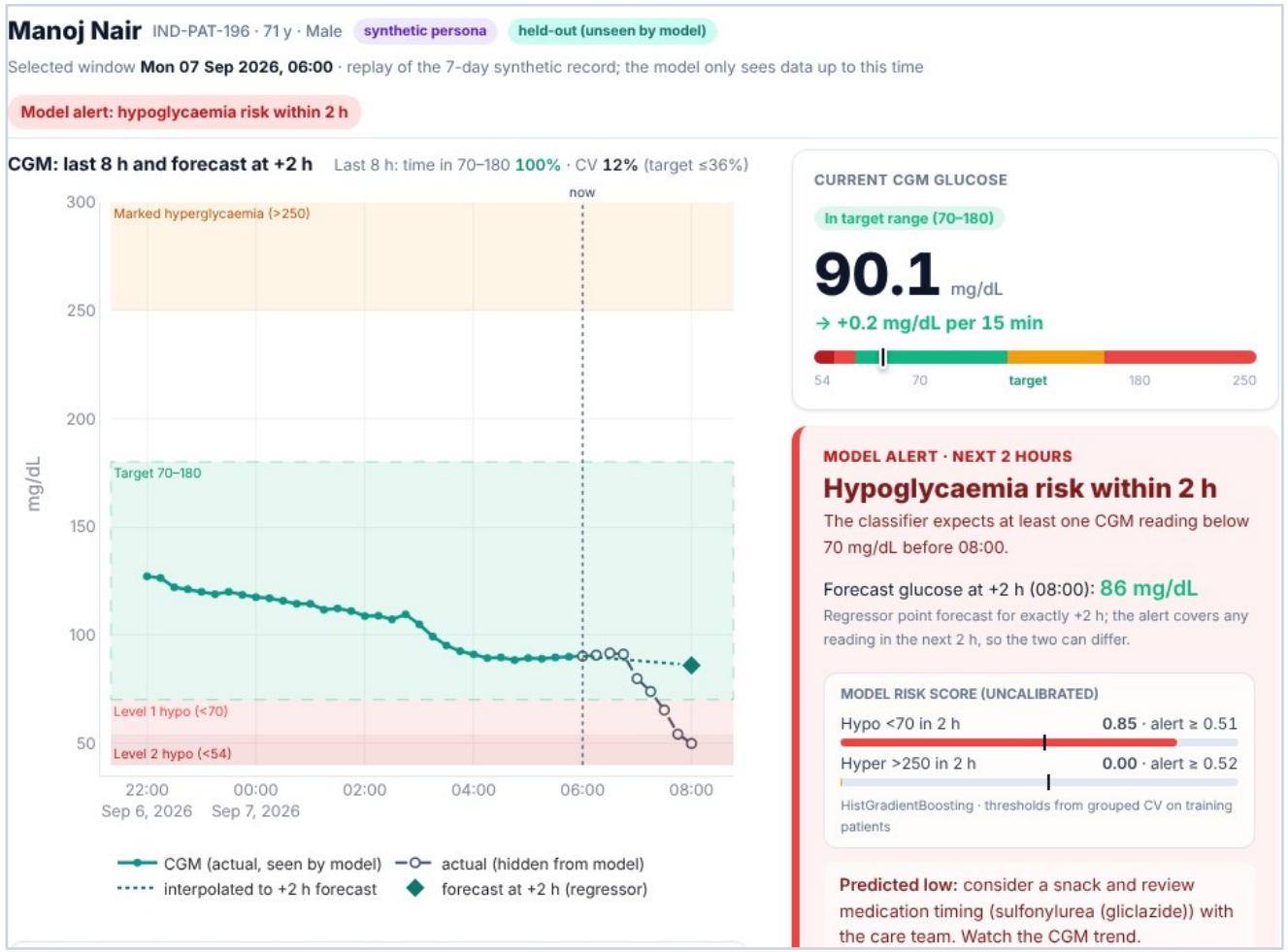
Held-out patients scored with another patient's EHR:

Forecast MAE **11.0 → 16.13 mg/dL**
Hypo PR-AUC **0.621 → 0.435**
Hyper PR-AUC **0.774 → 0.646**

- **The model uses the EHR:** the wrong record makes every score worse.
- **Small gains over wearables only:** MAE 11.45 → 11.08, hypo PR-AUC 0.530 → 0.548; hyper PR-AUC is higher without the EHR (0.855 vs 0.820).
- **No measurable signal** from HR, HRV and sleep (MAE 11.07 without them) or from history and genetics (hypo PR-AUC 0.555 without them).
- **One CV run,** no per-fold spread reported: gaps this small may be noise.

Source: models/metrics.json · Synthetic data only · research prototype, not a medical device · not for dosing decisions

The live monitor scores a held-out patient window by window, on a laptop or a phone

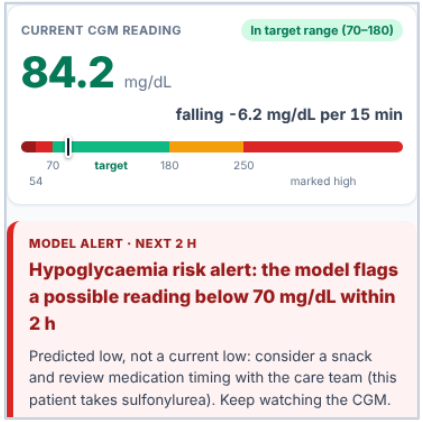


What the doctor sees

- Last 8 h of CGM, the +2 h forecast and the alert, with both uncalibrated risk scores and their thresholds
- Held-out patients: the readings that followed (hidden from the model) shown for comparison
- Predicted low: “consider a snack / review medication timing with the care team”. Rescue-carb text only when CGM is <70 now
- Scored live on CPU for each selected window

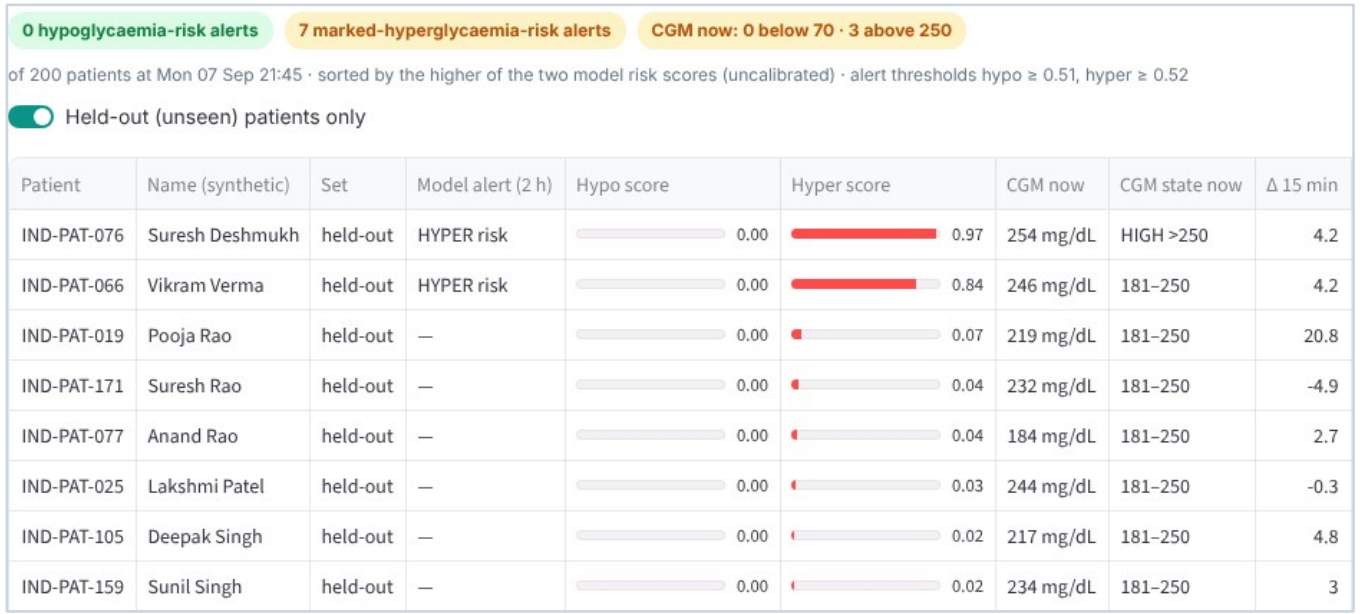
Hosted page on a phone

Same patient, an earlier alert. Outputs precomputed in Python; no model runs in the browser.



Source: dashboard/app.py, dashboard/index.html (screenshots) · Synthetic data only · research prototype, not a medical device · not for dosing decisions

A cohort risk board shows who needs attention; a rule-based sandbox supports counselling



0 hypoglycaemia-risk alerts | 7 marked-hyperglycaemia-risk alerts | CGM now: 0 below 70 · 3 above 250

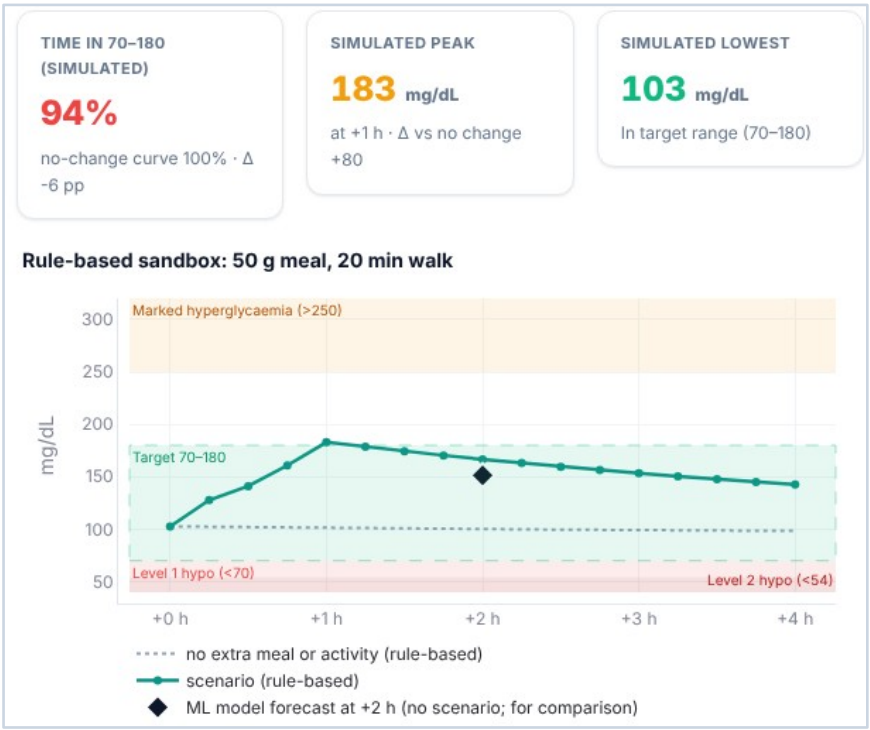
of 200 patients at Mon 07 Sep 21:45 · sorted by the higher of the two model risk scores (uncalibrated) · alert thresholds hypo ≥ 0.51, hyper ≥ 0.52

Held-out (unseen) patients only

Patient	Name (synthetic)	Set	Model alert (2 h)	Hypo score	Hyper score	CGM now	CGM state now	Δ 15 min
IND-PAT-076	Suresh Deshmukh	held-out	HYPER risk	0.00	0.97	254 mg/dL	HIGH >250	4.2
IND-PAT-066	Vikram Verma	held-out	HYPER risk	0.00	0.84	246 mg/dL	181–250	4.2
IND-PAT-019	Pooja Rao	held-out	—	0.00	0.07	219 mg/dL	181–250	20.8
IND-PAT-171	Suresh Rao	held-out	—	0.00	0.04	232 mg/dL	181–250	-4.9
IND-PAT-077	Anand Rao	held-out	—	0.00	0.04	184 mg/dL	181–250	2.7
IND-PAT-025	Lakshmi Patel	held-out	—	0.00	0.03	244 mg/dL	181–250	-0.3
IND-PAT-105	Deepak Singh	held-out	—	0.00	0.02	217 mg/dL	181–250	4.8
IND-PAT-159	Sunil Singh	held-out	—	0.00	0.02	234 mg/dL	181–250	3

Cohort risk board

Every patient scored in one batched call and ranked by the higher of the two risk scores (held-out patients shown)



What-if sandbox

Hand-written rules for counselling, not the ML model (its +2 h forecast is shown only for comparison); not a dosing tool

Source: dashboard/app.py (screenshots) · Synthetic data only · research prototype, not a medical device · not for dosing decisions

LIMITATIONS AND SAFETY

What this prototype does not show yet

- **Synthetic data only.** The model learns our simulator, which is simpler than real physiology; high ROC-AUC partly reflects that. No result transfers to real patients yet.
- **Uncalibrated simulator and scores.** Simulator parameters are hand-set; classifier outputs are risk scores, meaningful only against the thresholds.
- **False alarms.** 0.36 hypo and 0.58 hyper alarms per patient-day; 69% and 88% are not followed by the event.
- **Harder case: in range now.** If CGM is 70–250 at the time of the alert, sensitivity is 63.2% (hypo) and 77.9% (hyper).
- **Thin coverage.** 3 of 200 personas use insulin (0 in the held-out set); none has eGFR <60; only 35 readings <54, too few to evaluate Level 2 hypoglycaemia.
- **Regular routines.** Hour of day is a strong feature (MAE 13.86 without it); real days vary more.

Safety statement

Synthetic data only. Research prototype, not a medical device. Not for diagnosis or dosing decisions.

- Decision support for a clinician, who decides any action
- A predicted low suggests a snack or a medication-timing review with the care team
- Rescue text (15 g fast carbs, recheck in 15 min) only when CGM is <70 now; never insulin doses
- Uses only synthetic data, so no personal health data is processed (designed with DPDP Act 2023 principles in mind)
- Clinical use would need prospective validation and regulatory review

Source: data/synthetic_ehr.csv, models/metrics.json · Synthetic data only · research prototype, not a medical device · not for dosing decisions

NEXT STEPS

From a synthetic twin to a clinic pilot

1. **Real CGM data**
   Validate on open Type 2 diabetes CGM datasets (for example ShanghaiT2DM, CGMacros on PhysioNet) after licence checks
2. **Realistic EHR**
   Synthea with an India-adapted module; MIMIC-IV only where its data use agreement allows
3. **Calibration and alarms**
   Calibrate risk scores on grouped CV; tune thresholds with clinicians to cut false alarms
4. **Prospective pilot**
   Consented pilot with an Indian diabetes clinic; measure outreach and out-of-range time against usual care

Thank you

Code and docs
github.com/kumarshubhammorya/glycotwin-digital-twin

Hosted page
glycotwin-app.web.app

Video walkthrough (20+ min)
linked in README.md, section 5

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